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A genetic algorithm-based approach for job shop scheduling

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Abstract

Purpose – The purpose of this paper is to optimise the job shop scheduling problem using simulation and genetic algorithm.

Design/methodology/approach – The paper presents a simulation-based genetic algorithm approach for the job shop scheduling problem. In total, three cases have been considered to access the performance of the job shop, with an objective to minimise mean tardiness and makespan. A restart scheme is embedded into regular genetic algorithm in order to avoid premature convergence.

Findings – Simulation-based genetic algorithm can be used for job shop scheduling problems. Moreover, a restart scheme embedded into a regular genetic algorithm results in improvement in the fitness value. Single process plans selected on the basis of minimum production time criterion results in improved shop performance, as compared to single process plans selected randomly. Moreover, availability of multiple process plans during scheduling improves system performance measures.

Originality/value – The paper presents a simulation-based genetic algorithm approach for job shop scheduling problem, with and without restart scheme. In this paper the effect of multiple process plans over single process plans, as well as criterion for selection of single process plans, are studied. The findings should be taken into account while designing scheduling systems for job shop environments.

Keywords Genetic algorithms, Simulation, Production scheduling, Job shop scheduling

Paper type Research paper

1. Introduction

Scheduling problems are quite common occurrences in real world. It exists whenever there is a choice as to the order in which a number of tasks can be performed (Conway *et al.*, 1967). Generally, scheduling can be described as the allocation of a set of resources in a period of time, to perform a set of tasks. Resources may be machines in a shop floor, runways in an airport, crews at a construction site or processing units in a computing environment. Tasks may be operations in a shop floor, takeoffs and landing in an airport, stages in a construction project or computer programs to be executed (Vinod and Sridharan, 2008). Thus, scheduling is a decision-making process which can help to improves productivity of system (Ponnambalam *et al.*, 2001). In manufacturing, the scheduling depends upon the shop floor environment such as flow shop, job shop and open shop. Job shop scheduling (JSS) is the most typical class of scheduling. Analysis of JSS problem provides important insight into the solution of the scheduling problems encountered in more realistic and complicated system (Kutanoglu and Sabuncuoglu, 1999). Job shop performs "n" number of tasks on "m" number of machines. In a job shop, products are made to order in low volume. The work flow in a job shop is not unidirectional, each machine in the shop can be characterized by the input and

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output flows of work. JSS problem is well known as one of the most difficult NP-hard combinational optimisation problem (Pinedo, 1995). Hence, heuristics are preferred for JSS (Maccarthy and Liu, 1993). Genetic algorithm (GA) is the best-known optimising ability for a class of combinatorial problems (Wang and Zheng, 2002; Jia *et al.*, 2007; Buzatu and Bancila, 2008).

In this paper, a simulation-based GA approach is used for JSS problem. The evaluation of fitness function is carried out using simulation. Simulation is utilised to compute performance measures as it yields performance measure closed to actual system performance in comparison to mathematical functions. Mean tardiness and makespan are considered as performance measure.

This paper is organised as follows. Section 2 provides review of literature on JSS. Section 3 presents the details of adopted methodology. Section 4 gives the descriptions of results and finally conclusions from present work are drawn in Section 5.

2. Literature review

JSS represents several real situations of planning and for that many efforts have been made to investigate an effective implementation, along the past decade, particularly with GA. Davis (1985) was first to use GA to solve JSS problem. In literature, two important issues have been studied to solve JSS problem utilising GA's. One is how to encode (representation) a solution of the JSS problem into a chromosome so as to ensure that a chromosome will correspond to a feasible solution. In this direction a brief survey of existing representation schemes can be found in Amirthagadeswaran and Arunachalam (2006). Also, they proposed a method of representation and schedule deduction method utilising a heuristic technique in order to minimise makespan. Li and Chen (2010) designed a two-row chromosome structure based on working procedure and machine distribution in order to minimise makespan. Another issue is how to enhance performance of genetic search (local) by incorporating traditional heuristic methods. Moreover, JSS is a hard problem that cannot be solved effectively by applying any single technique and a great deal of research has focussed on various methods of hybridisation (Cheng *et al.*, 1999). Lee *et al.* (1997) proposed a framework utilising GA with machine learning and heuristics-space, in order to determine the sequence of job release order as well as the sequence for dispatching jobs at each individual machine. Yamada and Nakano (1997) proposed a multi-step crossover fusion (MSXF) as a unified operator and a recombination operator in GA local search for JSS, in order to solve large-size problems only. Ghedjati (1999) proposed a heuristics mixing method based on GA to solve JSS problem with consideration of parallel machines and precedence constraints in order to minimise the maximum completion time. Author concluded that proposed method obtained improvement due to heuristics but with longer running time. In Xie (2001) the job batch size and operation sequences were considered as two variables in order to achieve optimal job completion time. Authors concluded that job completion time could be reduced if batch size is considered. Goncalves *et al.* (2002) combines a GA to define the priority of operations and delay time, a schedule generator procedure to generate parameterized active scheduled, and a local search heuristics procedure to improve the solution. Wang and Zheng (2002) proposed a modified GA (MGA) for JSS using operation-based representation and decoding the solution into an active schedule during the search process, replacing the classical mutation of GA by the metropolis sample process of simulated annealing (SA) with probabilistic jumping property, applying

multiple state generators in a hybrid way. Mattfeld and Bierwirth (2004) considered release and due-dates in order to measure tardiness. Authors concluded that the weakness of GA to search for increasing problem size could be mitigated by a tuneable schedule builder and multi-stage decomposition technique which reduces the size of the search space. Hasan *et al.* (2007) combines a heuristics job ordering with GA for small to medium size JSS problems in order to find a valid schedule that yields the minimum makespan. Buzatu and Bancila (2008) integrate GA and combinations of heuristic rules, such as shortest processing time (SPT), minimal slack (MS), longest processing time (LPT) in order to generate the initial individuals. Zhou *et al.* (2009) proposed a hybrid framework in which GA determined first operation of each machine while heuristics determined the assignment of remaining operations in order to minimise weighted tardiness. Jia *et al.* (2007) integrate GA and Gantt chart (GC) for JSS in order to determine combination of process plans as well as operation schedule for small-sized or medium-size scheduling problems for a distributed manufacturing system. Some studies have been attempted for designing multi objective GA (MOGA) for JSS problem (Ponnambalam *et al.*, 2001).

Jain and Meeran (1999) provides a comprehensive survey of methods by which JSS problem has been approached. They concluded that meta-heuristic technique appears to work best for JSS problem. Among the various approaches to JSS problem, there has been an increasing interest in applying GA. A GA exhibits parallelism, contains certain redundancy and historical information of past solutions, and is suitable for implementation on massively parallel architectures (Wang and Zheng, 2002). The combined use of a GA and a heuristic is a typical form of meta-heuristics (Zhou *et al.*, 2009).

Literature reveals that numerous GA approaches to solve JSS problem have been utilised. These approaches differ strongly from each other with respect to the encoding and operators used, the constraints handled and the goals pursued. Despite these differences, all approaches have in common knowledge is required in order to produce competitive schedules. GA has been approved as an effective and efficient optimisation tool for JSS problems. Literature review divulges that simulation-based GA is not studied deeply for JSS problem. Whereas, simulation provides a meaningful understanding of real world system's nature. Thus, a simulation-based GA can be settled as an emphatic approach to accomplish JSS. The JSS problem can be defined as: there are "n" numbers of jobs that have to process on "m" numbers of machines in order to find an optimal job schedule. Each job "i" is to be processed on a set of machines and the processing time of each operation is known in advance. The resulting schedule is subjected to the constraint of job and machines, i.e. one machine can process only one operation at a time and pre-emption of any operation is prohibited.

3. Adopted methodology

The present work utilises a simulation-based GA approach. Encoding/representation is the first step of GA. A sequence oriented representation is used in the present work. Here, a bit (gene) of a chromosome is formed by a process plan number (i.e. alphabets) of a part type. Each bit of the chromosome is in fixed sequence/order to represent associate process plan of a part type. For example, if there are three part types A, B and C having of multiple process plans (MPP), respectively. Each part-type can be processed through any of its given MPP. Then, this information can be encoded as: $\{2, 1, 4\}$. Where, 2 represents

processing of part-type A by following its second process plan, 1 represents processing of part-type B by first process plan and 4 represents processing of part-type C by its fourth process plan. In this work, initial population is generated randomly. The present work utilise linear ranking (LR) selection method with stochastic universal sampling (SUS). In this method, individuals in the population are ranked according to their fitness and the expected value of each individual depends on its rank rather than on its absolute fitness (Baker, 1985). Once the expected values have assigned, the SUS method is applied to sample the population (i.e. choose parents). Thus, a mating pool consisting of selected individual is formed (Mitchell, 2002).

In this study, two-point crossover approach is considered and it is applied on the individuals of mating pool. For two-point crossover, two strings are selected randomly from the mating pool to make a pair. For each pair, essentiality of carrying out crossover is determined using crossover probability ($p_c = 0.8$). In two-point crossover site is selected randomly twice from first to last position. The present work utilise reciprocal exchange type of mutation operator with a mutation probability ($p_m = 0.2$) and it is applied on the off-springs produced after the crossover operation. The mutation position (from first to last bit) is selected randomly twice and process plans at these sites are interchanged, keeping other bits unchanged.

After mutation, some illegal off-spring may generate, i.e. one part type may exceeds the limit of given MPP in the individuals which undergoes mutation operation. Thus, a repairing strategy is required to resolve this illegitimacy of off-springs. In the present work, after mutation, all the individuals which undergoes mutation operation are checked to ensure that not any part type exceed the given range of process plans. Moreover, a repairing procedure also checks if at any position/site of string exceeds the specified number of concerned process plan of part type, it shall replaced by any one process plan from the given range of process plans randomly. For reproduction, the elitism way is embedded with the LR selection. Elitism transfer few good individuals from the previous population to the population of next generation. In the present work, an elitism rate of 0.9 is considered in order to transfer the best individual form pervious population to the population of next generation. In the present work maximum number of generation (*max_gen*) is considered as the termination criterion, i.e. equal to the number of jobs (n) $\times$ number of machines (m), as considered in Ponnambalam *et al.* (2001).

To overcome premature convergence in the population a restart scheme based on the ideas of a similar scheme used by Ruiz and Maroto (2006) is used. Accordingly, at each generation the maximum fitness value is store. If the best fitness value is not changed for more than a pre-specified number of generations (*best_count*), the restart phase commences to regenerate the population. In the present work, the value the counter (*best_count*) is set at 15, i.e. if the highest fitness value in the population does not change for more than 15 generations, the restart phase will be activate. This works as follows:

Step 1. Short the population in descending order of fitness value.

Step 2. Skip the first 20 percent of individuals from the shorted list.

Step 3. The remaining 80 percent of individuals in shorted list are disregarded and regenerated in the following way:

- Generate first half (50 percent) of new chromosomes by reciprocal exchange mutation operation of the first 20 percent best (shipped) individuals.
- Generate another half of new chromosomes randomly.

All newly generated genetic material will only replace the 80 percent worst individual of the population if they yield fitness value better than that of the worst individual of previous population. Also, the repetition of the chromosomes in the newly generated 80 percent population is not allowed.

The evaluation of fitness function is carried out using simulation. Simulation is used to compute performance measures as it yields performance measure close to actual system performance in comparison to mathematical functions. The present work considers the minimisation of mean tardiness of jobs and makespan. Figure 1 shows the flow chart of the working of GA.

In the present work, JSS environment for simulation can be characterized as follows: inter machine part transportation time is included/considered. Set-up time and inspection time are independent of schedule/sequence and are included in processing time. Infinite buffer location capacities are assumed in front of individual machine and each part enters in buffer location before the processing at machine. Ready time/release time of all jobs are zero, i.e. all jobs are available at the commencement of processing. Operation time and job arrival time are known and fixed before the job is released to shop. The routing of each job through the machine is random; for any given job, two consecutive operations on one machine are not allowed, but a job may return to that machine after being processed on other machines. The dispatching rule of SPT is used to process the jobs. First come first serve (FCFS) rule is used as a tie breaker. Due-dates are specified according to the total work content (TWK) criterion and MPP for each part type are considered. Machines are independent from each other and are available at zero time. Jobs are independent from each other. A part leaves the shop after all its operations are completed. Order cancellation, rework and machine breakdowns are not permitted.

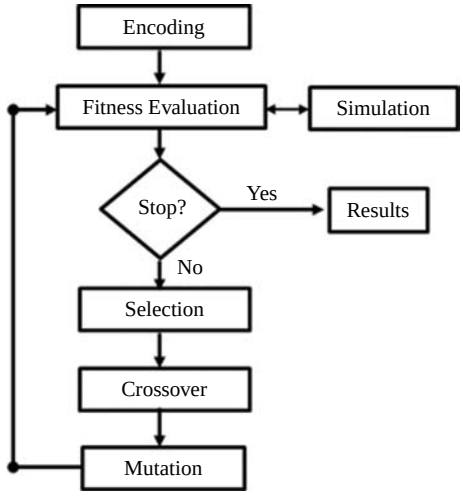


Figure 1.
Flow chart of working of genetic algorithm

4. Results and discussion

In order to implement adopted methodology, a job shop consisting of 15 machines is taken into consideration. Table I shows the details of a case study used for implementation. Uniform transportation time of three units is assumed between two machines.

Three cases are considers in order to measure the comparative performance of job shop. In all cases, mean tardiness and makespan are regarded as the performance measures. First case consists of randomly selected single process plan for each part type of production order. In second case, single process plan for each part type is considered according to minimum production time criterion. Third case deals with MPP for each part type as reported in production order.

These three cases are optimised using adopted methodology and Table II presents the results. It clearly reveals the randomly selected single process plan performs worst for

PT	Q	DD	PP no.	MPP's {Mi (Ti)}	TM	TT	TP
A	20	1000	1	14(10)-15(61)-12(43)-11(39)-8(47)	200	12	212
			2	14(10)-15(18)-13(30)-4(32)-11(39)-8(47)	176	15	191
B	15	455	1	5(10)-9(10)-1(33)-5(38)	91	9	100
			2	5(10)-8(6)-4(40)-5(38)	94	9	103
			3	6(25)-10(41)-14(24)-6(38)-10(37)	165	12	177
			4	6(25)-10(41)-13(12)-12(30)-7(39)-10(37)	184	15	199
C	18	345	1	4(29)-2(19)-7(21)	69	6	75
			2	4(29)-8(29)-6(9)-7(13)	80	9	89
			3	5(39)-14(33)-2(11)-1(10)-12(16)	109	12	121
			4	5(39)-1(6)-5(30)-3(39)-2(29)-1(19)	162	15	177
			5	10(69)-13(33)-8(29)	131	9	140
J	22	265	1	4(33)-15(20)	53	3	56
			2	1(22)-13(20)-7(35)-9(31)-12(45)-14(17)	170	15	185
			3	1(22)-12(76)-9(19)-3(48)-14(17)	182	12	194
K	11	335	1	10(27)-1(40)	67	3	70
			2	2(6)-13(36)	42	3	45
			3	6(30)-5(73)-4(31)	134	6	140
L	14	670	1	11(29)-15(44)-5(5)-12(41)-11(15)	134	12	146
			2	11(29)-15(24)-2(42)-11(15)	110	9	119
			3	3(22)-1(5)-9(39)-13(19)	85	9	94
			4	3(18)-6(5)-4(18)-7(39)-10(7)	87	12	99
			5	3(22)-7(10)-12(38)-13(19)	89	9	98

Table I.
Production order details

Notes: PT – part-type; Q. – production quantity; DD – due-dates; PP no. – process plan number; MPP – multiple process plans; TM – total machining time; TT – total transportation time; Mi (Ti) – machining number (processing time); TT – total production time

Performance measures	Single process plan (randomly selected)	Single process plan (selected with minimum production time criterion)	MPP
Mean tardiness	651	413	263
Makespan	2,292	2,268	1,541

Table II.
Results of case studies

Note: MPP – multiple process plans

both makespan and mean tardiness performance measures and MPP case yields least makespan and mean tardiness. Thus, it can safely be concluded that if only single process plan is to be made available than process plan selected according to minimum production time criterion yields better results than randomly selected process plans. Moreover, availability of MPP assists scheduler in selecting right combination of process plan for different part type resulting in the least makespan and mean tardiness.

This work also considers the suitability of restart scheme. Figure 2 shows the convergence curve of regular GA without restart scheme when MPP for each part type are available. The average fitness value obtained from regular GA is 0.00296. Figure 3 shows the convergence curve of GA with restart scheme for the above mentioned case. The average fitness value comes out to be 0.00331. Thus, embedding results scheme in regular GA avoids premature convergence and improves fitness value.

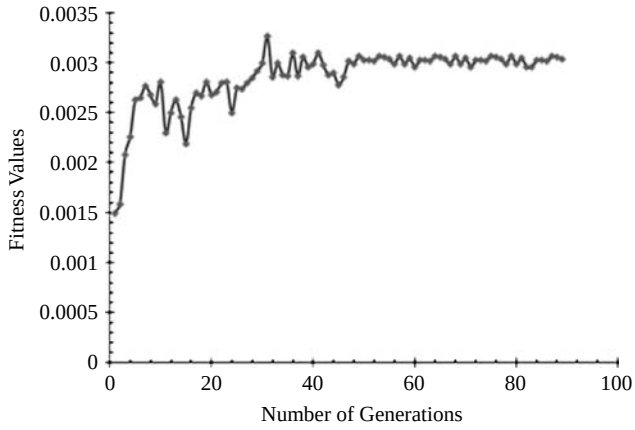


Figure 2.
Convergence curve of regular GA (without restart scheme)

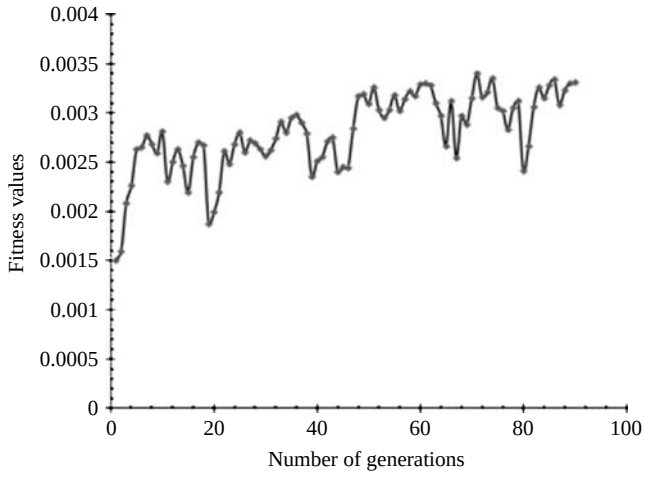


Figure 3.
Convergence curve with restart scheme

5. Conclusions

A simulation-based GA approach has been applied effectively to solve JSS problem. Mean tardiness and makespan are regarded as performance measures. Results indicates that the consideration of MPP for each part type have superior performance than single process plan for each part type. The comparative analysis shows that the embedding restart scheme into regular GA improves the fitness value of fitness function.

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